

Austin Senna Wijaya

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EDUCATION

Columbia University

Bachelor of Science: Computer Science

New York, NY

Aug 2024 – May 2028

- GPA: 4.22; Full-Tuition Scholarship; Dean's List.
- Research interests: data systems, AI agents, retrieval systems.
- Coursework: Natural Language Processing, Data Structures & Algorithms, Systems Programming, Linear Algebra & Probability, Competitive Programming.

RESEARCH EXPERIENCE

Data Agents Research Assistant

Dec 2025 – Present

Columbia Data, Agents, and Processes Lab

New York, NY

- Optimized core evaluation infrastructure for question-answering agents over a **9.5 TB/ 40M**-document data lake, including benchmark evaluation with parallel sandboxed workers, tool-call and reasoning telemetry, and BM25 and hybrid search engines.
- Improved **GPT-5-mini** semantic match from **2.22%** to **56.3%** by adding context compaction, loop-detection plugins, structured search-result context, and stronger data-analysis tools.
- Built an ablation framework to isolate where agents fail on question-answering on grounded reasoning tasks over long context windows through idealized search, planning, and data-analysis tools.

Machine Learning Research Assistant

Dec 2025 – Jun 2026

Columbia Zuckerman Institute

New York, NY

- Compressed **Qwen2-Audio 4096-D** hidden states into **10-100D** feature-specific representations, each preserving one linguistic or paralinguistic feature such as emotion or reasoning while suppressing other features.
- Curated training-ready datasets from **12** speech and language benchmarks, including converting text benchmarks into audio through parallel **Kokoro-TTS** generation.
- Built and parallelized a GPU-optimized **PyTorch** pipeline for Qwen2-Audio inference and hidden-state extraction, bottleneck-width sweeps, and supervised classification under cross-validation across **12** speech attributes and **120K+** audios.

PUBLICATIONS

- [1] *LakeQA: A Benchmark for Complex Exploratory QA over a Million-Scale Data Lake*, **ICML 2026**: Haonan Wang, Jiaxiang Liu, Yurong Liu, **Austin Senna Wijaya**, Tianle Zhou, Eden Wu, Yijia Chen, Wanting You, Reya Vir, Daniela Pinto Veizaga, Grace Fan, Yusen Zhang, Juliana Freire, Eugene Wu.
- [2] *SANA: What Matters for QA Agents over Massive Data Lakes?* **VLDB DASHSys 2026**: **Austin Senna Wijaya**, Jiaxiang Liu, Haonan Wang, Eugene Wu.
- [3] *MINA: Linear Probes Reveal Coding-Sequence Family Signal in Frozen DNA Encoders*, **iscbAI'26**: **Austin Senna Wijaya**, Hayden Leung. (Under-review).
- [4] *Targeted Embeddings: An Interpretable Framework for Neural Speech Encoding with Large Audio-Language Models*, **NeurIPS 2026**: Linyang He, Ciro D. Randazzo, **Austin Senna Wijaya**, Gabriel Diaz, Yunyan Ouyang, Jingting Zhang, Jonathan Brennan, Nima Mesgarani. (Under-review).
- [5] *Be CARE-ful with Text-to-SQL Benchmarks*, **NeurIPS 2026**: Haonan Wang, Jiaxiang Liu, Elaine Ang, Tiancheng Ge, Wanting You, **Austin Senna Wijaya**, Reya Vir, Tianle Zhou, Mihir Agarwal, Yusen Zhang, Eugene Wu. (Under-review).

INDUSTRY EXPERIENCE

Software Engineer Intern

Jun 2026 – Present

Garner Health

New York, NY

- Engineered a full-stack observability pipeline with automated dashboard and alert provisioning, implementing distributed tracing across Datadog RUM/Replay and OpenTelemetry (Loki, Prometheus, Tempo) to correlate frontend sessions with backend traces.
- Architected an autonomous on-call triage AI agent that ingests cross-stack telemetry to execute root-cause analysis and autonomously generate verifiable fix pull requests adhering to zero-PHI guardrails, resolving 10+ production incidents.

Data Engineering Intern

Jun 2025 – Aug 2025

Ruangguru (Indonesia's largest edtech platform)

Jakarta, Indonesia

- Consolidated millions of payment events into user sessions in **BigQuery**, producing a product funnel used to identify major checkout drop-off points.
- Automated Google Review scraping and sentiment analysis with **Puppeteer** and **Hugging Face**; processed tutor-scheduling data into live **Looker Studio** heatmaps for operations teams.

PROJECTS

ResearcherX: Local-First AI Research IDE

[Website](#)

- Built ResearcherX, a local-first AI research editor that helps researchers write against a bounded source library, using graph-based RAG to produce cited answers and detect unsupported claims, contradictions, or citation drift in drafts.
- Implemented the retrieval and verification pipeline behind the editor, turning PDFs/notes/links into provenance-tracked source nodes and checking draft paragraphs against relevant evidence with batched LLM calls.

MINA: Model Interrogation of Nucleotide Architectures

[Preprint](#)

- Built MINA, a homology-aware probing benchmark for frozen DNA encoders, testing whether linear probes recover **5**-way protein-family labels and **1,536-D** GenePT text embeddings across **3,244** human genes.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++, Java, JavaScript/TypeScript
- **Libraries & Frameworks:** Strands Agents, Langchain, Pandas, NumPy, PyTorch, Puppeteer/Selenium, BeautifulSoup, React
- **Cloud & Databases:** AWS (EC2, S3, Bedrock), Google Cloud (BigQuery, Looker Studio), Supabase, PostgreSQL, MongoDB
- **Tools:** GitHub, Claude Code, Codex, Docker, Copilot, Arduino, Notion

HONORS

- **Databricks Grounded Reasoning Cup 2026:** Columbia representative; worked with Google.
- **Venture Competitions 2026:** 2nd Place at Columbia Lion Cage; Top 6 at Millard Chan Technology Startup Competition.
- **Hack@Brown 2026:** Finalist; Strongest Product Thinking award; worked with Google Ventures and Partiful.
- **Jane Street Mystery Planet Trading & Puzzle Competition 2026:** 1st Place.
- **Tencent Kepler Plan S3 Competition 2025:** 2nd Place; Best Code; Best Product Idea; Popularity Award.
- **Indonesia Maju Scholarship:** Full-ride scholarship recipient, 1 of 350 recipients.
- **55th International Chemistry Olympiad 2023:** Final stage, 10th rank for Indonesian national team selection.